Advancing Sensor Network Optimization and Data Analytics for Smart Wastewater and Stormwater Collection Systems: An Overview

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ABSTRACT

As the foundation of monitoring systems, knowledge of the stormwater and wastewater collection systems' performance is essential to preserving structural integrity, averting system failure, and guaranteeing their dependable and effective daily operation under typical circumstances. Nevertheless, these subterranean systems encounter major obstacles because of restricted access, occasional examinations, and aging-related deterioration, such as corrosion, obstruction, and fouling, in addition to high demands brought on by population growth and the constant rise in stormwater demand brought on by flooding. Real-time monitoring, predictive maintenance, and consequently prompt decision-making are made possible by recent developments in sensing, artificial intelligence (AI), machine learning (ML), and decision support systems. In keeping with this, this paper provides a summary of the most recent developments in sensor network optimization by incorporating AI-driven data analytics and decision tools to enhance operational efficiency, infrastructure resilience, and data processing. To discover how different optimization techniques, such as heuristics, metaheuristics, and reinforcement learning, can improve sensor placement strategies and strike a compromise between cost limitations and data observability, they are compiled here. The advantages of AI-based predictive analytics in enhancing wastewater monitoring, cutting maintenance expenses, and enhancing regulatory compliance are illustrated by case examples.

INTRODUCTION

Outdated techniques, infrequent inspections, and fragmented data processing make managing wastewater and stormwater collection systems impossible, resulting in inefficiencies and regulatory issues (Tariq et al., 2021). The necessity for data-driven solutions that improve system monitoring and maintenance is highlighted by the fact that aging infrastructure, the effects of climate change, and mounting environmental demands all increase operational risks (Liggett et al., 2018; Salveson, 2023).

Real-time monitoring and predictive maintenance are made possible by recent developments in sensor networks and data analytics, which present potential possibilities. Preventive maintenance, capital planning, and hydraulic performance are all enhanced by strategically placed sensors, such as flow, pressure, and rain gauges (Bersinger et al., 2015). However, widespread implementation is hampered by practical issues such debris accumulation, sensor fouling, and expensive maintenance costs (Steinle-Darling et al., 2020). Machine learning (ML) used with conventional monitoring systems improves predictive capabilities, increasing efficiency and dependability to overcome these issues (Kumar et al., 2024).

To promote the development of smart sewer systems, real-time sensor technology is also essential for the identification of contaminants and biomarkers (Nourinejad et al., 2021; Sambito et al., 2020). Optimal sensor placement is a crucial area of research since, despite these developments, high installation and maintenance costs continue to be major obstacles (Banik et al., 2015). To optimize system insights, sensor network design must strike a compromise between cost considerations and data collecting effectiveness (Banik et al., 2017; Salem & Abokifa, 2024). New technologies like fiber optics and Micro-Electro-Mechanical Systems (MEMS) improve scalability and lower prices while increasing real-time data gathering and transmission (Geetha & Gouthami, 2016; Tariq, Hu, & Zayed, 2021). Low-power, long-distance data transport is made possible by fiber optics, which increases network resilience, while MEMS-based sensors allow for cost-effective deployment over dispersed networks. Furthermore, by guaranteeing regulatory compliance, enhancing resource management, and enabling near real-time monitoring, smart metering has revolutionized utility operations (Kumar et al., 2024).

Nonetheless, adoption is still being slowed by industry reluctance brought on by worries about costs and performance dependability in challenging wastewater conditions (Liggett, Macintosh, & Thompson, 2018). The shift from disparate datasets to coherent, actionable intelligence is further supported by developments in cloud computing and big data analytics (Salveson, 2023). Utility companies can improve maintenance plans, avoid system breakdowns, and optimize flow regulation by utilizing AI-driven predictive analytics (Steinle-Darling et al., 2022). To fully utilize smart wastewater systems, however, enduring issues with sensor performance, power supply, and signal processing must be resolved (Tariq, Hu, & Zayed, 2021).

This paper closes this gap by offering a thorough review of sensor network advancements that integrate data analytics and AI-driven decision-making with applications in utility system efficiency enhancement and sensing placement. Using case studies, this overview will highlight how AI/ML tools could improve data gathering, optimize sensor placement, and transform the data into useful insights.

OPTIMIZATION OF SENSOR NETWORKS

Advanced sensor placement techniques are necessary for wastewater collecting networks to improve efficiency through optimum monitoring procedures. Sensor placement places a higher priority on data observability, contamination detection, and coverage than previous network design, which was more concerned with flow efficiency and cost. By combining hydraulic simulations with predictive analytics, AI-driven optimization, including deep learning and reinforcement learning—has enhanced sensor deployment (Zhao et al., 2016). There are several types of optimization techniques, including gradient-based, heuristic, metaheuristic, and precise methods. Although exact methods like linear and convex optimization (Zhao et al., 2016) yield accurate results, they are only applicable in certain situations. Heuristic techniques, such as greedy algorithms (Banik et al., 2017), are effective but might not produce the best results. Genetic algorithms and other metaheuristic approaches allow for more thorough investigation but come with a larger computing cost. Large-scale optimization is enhanced by gradient-based techniques (Jafari, Nazif, and Rajaee, 2022), albeit they are hyperparameter-sensitive.

Utilities prioritize sensor placement in high-risk areas based on previous failures due to budgetary constraints (Haby et al., 2015; Kimbrough, 2019). Monitoring programs that target frequent spill sites have been effectively implemented by cities such as Atlanta and San Antonio (Fenner & Sweeting, 1999; Macrina & Woodall, 2016; Haby, 2013). But these tactics might miss unreported overflows, necessitating more advanced optimization methods (Kimbrough, 2019). The placement of wastewater sensors is informed by lessons learned from related applications. While hydraulic models maximize leak detection in water distribution networks (Krause et al., 2008; Yassine et al., 2008), wireless sensors identify structural degradation in bridge monitoring (Lynch, 2007). However, because of regional variability, its direct application to wastewater systems is still difficult (Kimbrough, 2019).

Sensor optimization has made use of metaheuristic techniques such as Greedy Randomized Adaptive Search Procedures (GRASP), Tabu Search, and Simulated Annealing (Baghel et al., 2012). Although Lin et al. (2005) showed that simulated annealing works well in grid-based layouts, infrastructural constraints like manhole location and spatial autocorrelation limit its practical use in wastewater systems (Kimbrough, 2019). Sensor optimization is further refined using objective functions. Specific monitoring objectives are addressed by metrics including the Fisher Information Matrix (FIM), Kinetic Energy (KE), Effective Independence-Point Residue (EI-DPR), Effective Independence (EI), and Modal Assurance Criterion (MAC). Furthermore, computational complexity is introduced by multi-objective optimization, which balances contamination detection with cost limitations (Watson, Greenberg, and Hart, 2004). More flexible approaches run the danger of producing suboptimal layouts, whereas convex optimization may induce epistemic uncertainty. For wastewater monitoring to be effective, a customized strategy combining sensor applications, optimization techniques, and objective functions is necessary.

AI/ML BASED SENSOR PLACEMENT

Growth in data analytics has been made possible by the development of artificial intelligence (AI) and machine learning (ML) techniques, which allow for learning from data and information extraction from data from a statistical perspective. Thanks to developments in sensing and computing technology, new computation intelligence with potential applications in water utility domains is becoming rapidly available. Wastewater collection systems employ classification and predictive analytics using decision trees and other machine learning techniques. Conditional assessment, threat detection, early-warning anomaly classification, and maintenance strategy decisions are just a few of the water utility applications for which these data analytics techniques can be wisely applied to data classification, data regression, or data clustering in supervised learning or unsupervised learning.

Table 1 highlights the decision tree as a conventional, straightforward, and interpretable model among the several machine learning techniques. It is extensively suitable in water utility applications because of its simple form, which makes it possible to provide concise, visual explanations of decision-making procedures. In water systems, decision trees have been successfully used for a variety of tasks, such as maintenance (Hidayat et al. 2023), risk assessment (Yang, Hu, and Zheng 2020b, 2020a), deterioration prediction (Dawood et al. 2020), and damage detection (Yang, Hu, and Zheng 2020a). Decision trees are still a useful tool for solving more complicated water utility issues with a high degree of uncertainty. By combining many decision paths, this method efficiently reduces epistemic uncertainty (uncertainty resulting from ignorance) and aleatoric uncertainty (uncertainty inherent in the system) by creating a decision forest.

Table 1. Summary of Machine Learning Models in Sensor-Based Collection Systems

Technology	Applications	Benefits	Limitations
Decision tree	Can process many input features but are prone to overfitting if irrelevant features are included	Simple to interpret and visualize. Can handle both numerical and categorical data	Prone to overfitting, especially with high-dimensional data. Sensitive to small variations in the data.
Random forest	Can process high-dimensional data well due to their use of feature bagging	Reduces overfitting by averaging multiple decision trees. Provides feature importance scores, aiding in feature selection.	Less interpretable than a single decision tree. Computationally expensive due to the ensemble of trees.
Logistic Regression	Best with a moderate number of relevant features	Simple and interpretable model. Works well with linearly separable data	Limited in capturing complex relationships in high-dimensional spaces. Assumes linear relationships between features and the log-odds.
Support Vector Machines (SVM)	Work well with high-dimensional data but might become computationally expensive	Effective in high-dimensional spaces. Works well with a clear margin of separation.	Computationally intensive, especially with non-linear kernels. Not as interpretable
k-Nearest Neighbors (k-NN)	Effective in low dimensions, hindered by high dimensionality.	Simple and easy to implement. No need for a training phase.	Effective in low dimensions, hindered by high dimensionality.
Gaussian Process	Low dimension datasets	Provides uncertainty estimates for predictions. Flexible and non-parametric.	Computationally expensive with large datasets or high-dimensional data. Difficult to scale large datasets.
Neural Networks	Can handle very high-dimensional data, but require large amounts of data to avoid overfitting	Capable of capturing complex patterns and relationships. Highly flexible with different architectures.	Requires large amounts of data and computational resources. It is harder to interpret and tune.

Various identification and decision-making procedures in sensor gathering systems can be replaced by machine learning models, such as decision trees. The effectiveness of these systems, however, depends on selecting the best model for each stage. The most popular machine learning models have been outlined here, along with their advantages, disadvantages, and preferred applications, to help with the design and selection of sensor collection systems. Moreover, after an AI model is chosen for an existing system, its features play a crucial role in directing arrangements and sensor selection.

EXAMPLES OF SENSOR PLACEMENT STRATEGIES

Using case studies, this section will demonstrate how AI-driven analytics and strategically placed sensors improve wastewater system performance, facilitating operational resilience, regulatory compliance, and predictive maintenance. Table 2 highlights the influence of sensor network optimization across various utility systems by summarizing the main technologies, difficulties, results, and cost-effectiveness elements of these practical implementations.

Table 2 illustrates how United Utilities used the Dynamic Network Management (DNM) platform, which combines real-time monitoring and AI-driven analytics, to enhance their wastewater collecting system. The utility, which oversees 77,000 km of sewer networks, had to deal with obstructions and pump failures made worse by severe weather and a rise in the use of wet wipes. In 18 months, DNM generated 1,500 alarms and 550 actions by using the AquaDNA platform to analyze data from 1,980 lift stations and 20,000 sewer assets. This demonstrated the effect of AI-based sensor networks on predictive maintenance and system resilience by reducing maintenance calls by 25% and flooding events by 39% (Thompson et al., 2025). By incorporating AI-driven defect analysis, the City of Houston improved its pipeline inspection procedure. The city had to deal with inefficiencies in condition assessment while managing 6,500 miles of sewer mains under a consent decree. By increasing the number of pipes inspected by 72% and decreasing the review time per inspection by 80%, AI-based conduct closed-circuit television (CCTV) inspections and cloud-based data management improved accuracy while lowering staffing requirements and budget overruns.

To enhance its combined sewer system, the City of Omaha integrated Supervisory Control and Data Acquisition (SCADA) data, Geographic Information System (GIS) analytics, automated quality control, and real-time flow meters and rain gauges. According to Thompson et al. (2025), this digital dashboard enhanced regulatory compliance and response efficiency by improving infrastructure capacity evaluations and model refinement. Like this, the St. Louis Metropolitan Sewer District (MSD) used digital twin technology, AI-driven analytics, and real-time monitoring for wet weather operations. To minimize combined sewer overflows (CSOs) volume and avoid system surcharges, the Wet Weather Optimization Project (WWOP) optimized gate control tactics by integrating weather inputs and SCADA data into a predictive modeling dashboard. Early IT involvement and ongoing user feedback to improve system performance were important lessons learned (Thompson et al., 2025). Real-time monitoring was implemented by Los Angeles County Sanitation Districts (LACSD) to improve industrial compliance and effluent quality. With 11 treatment facilities serving 5.7 million people, LACSD created a pretreatment program that may identify pollution in as little as 24 hours. Targeted mobile sensors are being investigated for localized pollution tracking, while full-scale deployment is still expensive.

Table 2. Summary of Sensor Optimization Strategies

Case Study	Technologies Used	Main Challenges	Key Outcomes	Cost Efficiency
United Utilities	AI-driven predictive analytics, real-time monitoring, AquaDNA platform	Pump failures, blockages, extreme weather impacts	25% fewer maintenance calls, 39% fewer flooding incidents	Lower maintenance costs, reduced flood damage expenses
City of Houston	AI-based CCTV inspection, cloud-based data management	Manual inspection delays, data inaccuracy, compliance	80% faster review, 72% more pipes inspected, improved compliance	Reduced labor costs for inspections, improved resource allocation
City of Omaha	Real-time flow monitoring, GIS-integrated cloud dashboard	CSO compliance, data integration, infrastructure assessment	Faster decision-making, enhanced data accuracy, CSO compliance	Minimized regulatory penalties, optimized infrastructure investment
MSD St. Louis	AI-driven analytics, digital twin, SCADA integration	CSO management, wet weather optimization, system surcharges	Optimized CSO control, predictive management, improved resilience	Lower operational costs through AI-based optimization and automation
LACSD	Online monitoring sensors, automated sampling	Pollution detection, sensor placement, compliance tracking	Faster pollution detection, early compliance tracking	Reduced treatment plant upsets, lower compliance costs
Real-Time Collection System Monitoring (WRF 4908)	Real-time sensor network, pollutant tracking	Sensor accuracy, fouling, power limitations, data reliability	Improved pollutant tracking, industrial compliance monitoring	Avoided regulatory fines, reduced monitoring costs

The Water Research Foundation (WRF) 4908 study used pollution tracking probes in Ventura Water, El Paso Water, and Clean Water Services (CWS) to further illustrate the importance of real-time monitoring. The potential of improved sensor installation for compliance monitoring was highlighted by these utilities' identification of industrial pollution sources, which included oil refineries, cleaning products, and home water softeners. Notwithstanding obstacles including communication reliability, power supply constraints, and sensor fouling, the study reaffirmed the value of real-time monitoring in regulatory supervision (Steinle-Darling et al., 2020; Thompson et al., 2025).

It is obvious that maximizing sensor installation in wastewater networks necessitates striking a balance between financial restraints and technical viability. Financial constraints frequently limit sensor deployment to high-risk locations identified by historical data on blockages and overflows, even though complete system-wide monitoring is ideal (Haby et al., 2015; Kimbrough, 2019). The effectiveness of past failures as predictive indicators has been reinforced by the prioritization of monitoring in regions with recurrent events by cities like Atlanta and San Antonio (Fenner & Sweeting, 1999; Macrina & Woodall, 2016; Haby, 2013). This approach has drawbacks, though, as it might miss unreported overflows or sporadic obstructions (Kimbrough, 2019).

Sensor optimization techniques are further improved by insights from other infrastructure systems. Wireless sensors have been used in structural health monitoring to identify bridge degradation (Lynch, 2007), and hydraulic models in water distribution networks to identify leaks and contamination (Krause et al., 2008; Yassine et al., 2008). However, regional uncertainty and uneven model availability limit direct applicability to wastewater networks (Kimbrough, 2019). Though their application in wastewater systems is still restricted because of spatial autocorrelation and installation difficulties, advanced placement strategies rely on metaheuristic optimization techniques like Greedy Randomized Adaptive Search Procedures (GRASP), Tabu Search, and Simulated Annealing (Baghel et al., 2012; Lin et al., 2005). Because water quality monitors are expensive, deployment solutions that maximize efficiency while reducing costs are necessary (Salem & Abokifa, 2024). Numerous initiatives from the Water Research Foundation (WRF), such as WRF 4908, WRF 5048, WRF 4797A, and WRF 4797B, have shown the advantages of combining sensor technology with data analytics and intelligent platforms.

CONCLUSIONS

This report provides an overview of the most recent initiatives to optimize sensor networks and integrate AI-driven analytics to improve the stormwater and wastewater collection systems' resilience, efficiency, and affordability. Utility systems have effectively adopted a variety of sensor placement techniques, such as heuristic, metaheuristic, and reinforcement learning approaches, to enhance data observability and enable the resolution of financial and infrastructure restrictions. According to cited case studies, there are several advantages to using AI-based predictive analytics in practical applications, such as lower maintenance costs, better system performance, and increased regulatory compliance.

Furthermore, the current technical limits brought on by sensor fouling, power supply limitations, and industry adoption obstacles can be addressed thanks to developments in MEMS sensors, fiber optics, and cloud-based analytics. By improving the predictive power of wastewater monitoring systems using machine learning techniques like decision trees, random forests, and neural networks, utilities can transition from reactive to proactive management. Utilities may enhance flow regulation, avoid system breakdowns, and optimize maintenance plans by utilizing AI-driven decision-making. As seen by several case studies, the effective implementation of real-time monitoring systems highlights the significance of advanced data analytics and thoughtful sensor placement in wastewater management.

RECOMMENDATIONS FOR FUTURE RESEARCH

Future studies should concentrate on improving sensor network architectures and incorporating AI-powered predictive analytics for capital planning, preventive maintenance, and real-time flow control in collection systems. Utility decision-making regarding the effects of extreme weather and regulatory compliance will be improved by improving data analytics, forecasting, and intelligent dashboards. Interoperability between systems will be made possible by standardizing communication protocols and sensor data formats. Additionally, to safeguard smart monitoring infrastructure, cybersecurity measures need to be developed. System efficiency and scalability will be further enhanced by researching economical sensor deployment techniques, the robustness of sensors in a range of environments, and the use of cutting-edge technologies like fiber optics and MEMS-based sensors. Finally, studies should look into how AI adoption affects utility operational adaptation and workforce training.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the financial support provided by Water Research Foundation (WRF)-5239. The results, discussion, and opinions reflected in this paper are those of the authors only and do not necessarily represent those of the sponsor and participating utilities/organizations.

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